

A Brief Introduction to Computational Neuroscience

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1.0 Introduction

Computational neuroscience is a branch of neuroscience that uses mathematical and computational techniques to understand the brain. It is a relatively new field that combines neuroscience, computer science, and mathematics.

In fact, as you read the previous sentence or the upcoming words, your eyes are scanning across the screen in micro-jerking movements. Your eyes' tiny, precise muscles are controlled by your brain's **magnocellular tracking visual system**, which decodes individual letters and words (graphemes) into their representative sounds (phonemes). This will rapidly transform the squiggles that make up the word “book” into the sound for “book.”

Meanwhile, **Wernicke's area** enables you to comprehend the *meaning* of words. Your **angular gyrus** works hard to connect visual, auditory, semantic (meaning), and syntactic (grammar) information. And finally, **working memory** helps to retain what happens at the

beginning of this sentence up to the very end of the sentence. Did this frustrate you? Well, that's the **amygdala**.

Don't worry about these terms if you have never heard them before. We will cover them in the upcoming days.

This field is concerned with designing models or mapping neural networks and using these models to understand how the nervous system works. One of the goals of computational neuroscience is to develop a detailed, mathematical understanding of the brain.

1.1 Research fields

Some of the topics that computational neuroscience research covers include:

How do neurons compute?

How does the brain store and retrieve memories?

How does the brain control movement?

What are the neural mechanisms underlying perception?

What is the neural basis of consciousness?

2.0 Why combine computations and neuroscience?

There are many reasons why this field is important. First, the brain is the most complex organ in the body, and understanding how it works is a major challenge, and studying it from only a single perspective or field is like taking a restrictive approach. Second, the brain is responsible for everything we do, from the simplest reflexes to the highest level of cognition. Understanding the brain can help us better understand ourselves and find solutions to developmental issues and injuries related to the brain.

Third, the brain constantly changes in response to the environment and from experience. This makes it difficult to study using traditional methods. Computational neuroscience provides powerful tools for understanding the brain and studying how it changes over time.

Finally, combining data analysis and neuroscience is noteworthy because it has the potential to improve our lives. For example, by understanding how the nervous system works, we can develop better treatments for neurological disorders and conditions like Alzheimer's disease, Parkinson's disease, and stroke. We can also develop better prosthetic devices for people with spinal cord injuries.

3.0 Goals of Computational Neuroscience

One of the main goals of computational neuroscience is to understand how the brain works. Using mathematical and computational models, researchers can gain insights into the workings of the brain that would not be possible through traditional experimental methods.

One area where computational neuroscience has made great strides is in the area of brain-machine interfaces (BMIs). BMIs are devices that allow humans to control machines with their thoughts. Researchers have used BMIs to control robotic arms, computer keyboards, and video games. The success of BMIs is due, in part, to the fact that they take advantage of the brain's natural ability to learn and adapt. When people use a BMI, they are essentially training their brains to control the device. The more they use the BMI, the better their brain gets at controlling it.

This same principle can be applied to other research areas, such as understanding how the brain learns and remembers. By using computational models, researchers can better understand how the brain works and how to improve its function.